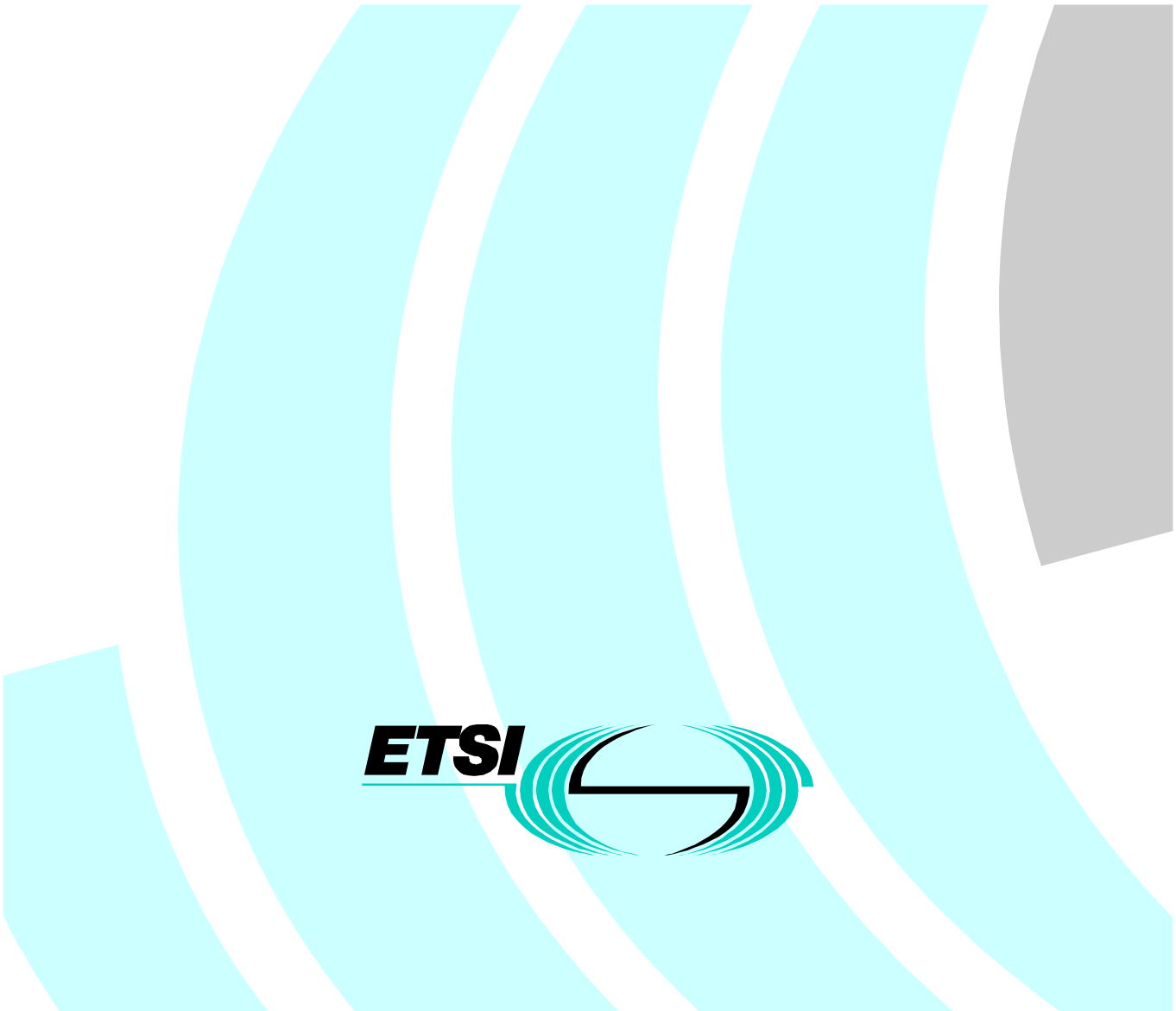
EN 300 359-5 V1.2.4 (1998-06)

European Standard (Telecommunications series)

Integrated Services Digital Network (ISDN); Completion of Calls to Busy Subscriber (CCBS) supplementary service; Digital Subscriber Signalling System No. one (DSS1) protocol; Part 5: Test Suite Structure and Test Purposes (TSS&TP) specification for the network



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Foreword

This European Standard (Telecommunications series) has been produced by ETSI Technical Committee Signalling Protocols and Switching (SPS).

The present document is part 5 of a multi-part standard covering the Digital Subscriber Signalling System No. one (DSS1) protocol specification for the Integrated Services Digital Network (ISDN) Completion of Calls to Busy Subscriber (CCBS) supplementary service, as described below:

Part 1: "Protocol specification";

Part 2: "Protocol Implementation Conformance Statement (PICS) proforma specification";

Part 3: "Test Suite Structure and Test Purposes (TSS&TP) specification for the user";

Part 4: "Abstract Test Suite (ATS) and partial Protocol Implementation eXtra Information for Testing (PIXIT) proforma specification for the user";

Part 5: "Test Suite Structure and Test Purposes (TSS&TP) specification for the network";

Part 6: "Abstract Test Suite (ATS) and partial Protocol Implementation eXtra Information for Testing (PIXIT) proforma specification for the network".

The present version updates the references to the basic call specifications.

National transposition dates	
Date of adoption of this EN:	19 June 1998
Date of latest announcement of this EN (doa):	30 September 1998
Date of latest publication of new National Standard or endorsement of this EN (dop/e):	31 March 1999
Date of withdrawal of any conflicting National Standard (dow):	31 March 1999

1 Scope

This fifth part of EN 300 359 specifies the Test Suite Structure and Test Purposes (TSS&TP) for the Network side of the T reference point or coincident S and T reference point (as defined in ITU-T Recommendation I.411 [7]) of implementations conforming to the stage three standard for the Completion of Calls to Busy Subscriber (CCBS) supplementary service for the pan-European Integrated Services Digital Network (ISDN) by means of the Digital Subscriber Signalling System No. one (DSS1) protocol, EN 300 359-1 [1].

A further part of this EN specifies the Abstract Test Suite (ATS) and partial Protocol Implementation eXtra Information for Testing (PIXIT) proforma based on the present document. Other parts specify the TSS&TP and the ATS and partial PIXIT proforma for the User side of the T reference point or coincident S and T reference point of implementations conforming to EN 300 359-1 [1].

2 Normative references

References may be made to:

a) specific versions of publications (identified by date of publication, edition number, version number, etc.), in which case, subsequent revisions to the referenced document do not apply; or

b) all versions up to and including the identified version (identified by "up to and including" before the version identity); or

c) all versions subsequent to and including the identified version (identified by "onwards" following the version identity); or

d) publications without mention of a specific version, in which case the latest version applies.

A non-specific reference to an ETS shall also be taken to refer to later versions published as an EN with the same number.

[1] EN 300 359-1 (V1.2): "Integrated Services Digital Network (ISDN); Completion of Calls to Busy Subscriber (CCBS) supplementary service; Digital Subscriber Signalling System No. one (DSS1) protocol; Part 1: Protocol specification".

[2] EN 300 359-2 (V1.2): "Integrated Services Digital Network (ISDN); Completion of Calls to Busy Subscriber (CCBS) supplementary service; Digital Subscriber Signalling System No. one (DSS1) protocol; Part 2: Protocol Implementation Conformance Statement (PICS) proforma specification".

[3] ISO/IEC 9646-1: "Information Technology - OSI Conformance Testing Methodology and Framework; Part 1: General Concepts".

[4] ISO/IEC 9646-2: "Information Technology - OSI Conformance Testing Methodology and Framework; Part 2: Abstract Test Suite specification".

[5] ISO/IEC 9646-3: "Information Technology - OSI Conformance Testing Methodology and Framework; Part 3: The Tree and Tabular Combined Notation".

[6] EN 300 196-1: "Integrated Services Digital Network (ISDN); Generic functional protocol for the support of supplementary services; Digital Subscriber Signalling System No. one (DSS1) protocol; Part 1: Protocol specification".

[7] ITU-T Recommendation I.411 (1993): "ISDN user-network interfaces - Reference configurations".

[8] EN 300 403-1: "Integrated Services Digital Network (ISDN); Digital Subscriber Signalling System No. one (DSS1) protocol; Signalling network layer for circuit-mode basic call control; Part 1: Protocol specification [ITU-T Recommendation Q.931 (1993), modified]".

[9] ITU-T Recommendation I.112: "Vocabulary and terms for ISDNs".

[10] CCITT Recommendation E.164: "Numbering plan for the ISDN era".

[11] ITU-T Recommendation I.210: "Principles of the telecommunication services supported by an ISDN and the means to describe them".

3 Definitions

For the purposes of the present document, the following definitions apply:

3.1 Definitions related to conformance testing

abstract test case: Refer to ISO/IEC 9646-1 [3].

Abstract Test Suite (ATS): Refer to ISO/IEC 9646-1 [3].

Implementation Under Test (IUT): Refer to ISO/IEC 9646-1 [3].

implicit send event: Refer to ISO/IEC 9646-3 [5].

lower tester: Refer to ISO/IEC 9646-1 [3].

point of control and observation: Refer to ISO/IEC 9646-1 [3].

Protocol Implementation Conformance Statement (PICS): Refer to ISO/IEC 9646-1 [3].

PICS proforma: Refer to ISO/IEC 9646-1 [3].

Protocol Implementation eXtra Information for Testing (PIXIT): Refer to ISO/IEC 9646-1 [3].

PIXIT proforma: Refer to ISO/IEC 9646-1 [3].

system under test: Refer to ISO/IEC 9646-1 [3].

Test Purpose (TP): Refer to ISO/IEC 9646-1 [3].

3.2 Definitions related to EN 300 359-1

Call Reference (CR): See EN 300 403-1 [8], subclause 4.3.

component: See EN 300 196-1 [6], subclause 11.2.2.1.

Integrated Services Digital Network (ISDN): See ITU-T Recommendation I.112 [9], definition 308.

ISDN number: A number conforming to the numbering and structure specified in CCITT Recommendation E.164 [10].

invoke component: See EN 300 196-1 [6], subclause 11.2.2.1.

network: The DSS1 protocol entity at the Network side of the user-network interface where a T reference point or coincident S and T reference point applies.

network (S/T): The DSS1 protocol entity at the Network side of the user-network interface where a coincident S and T reference point applies.

network (T): The DSS1 protocol entity at the Network side of the user-network interface where a T reference point applies (Network connected to Private ISDN).

return error component: See EN 300 196-1 [6], subclause 11.2.2.1.

return result component: See EN 300 196-1 [6], subclause 11.2.2.1.

served user: The served user is the user who invokes the CCBS supplementary service.

service; telecommunication service: See ITU-T Recommendation I.112 [9], definition 201.

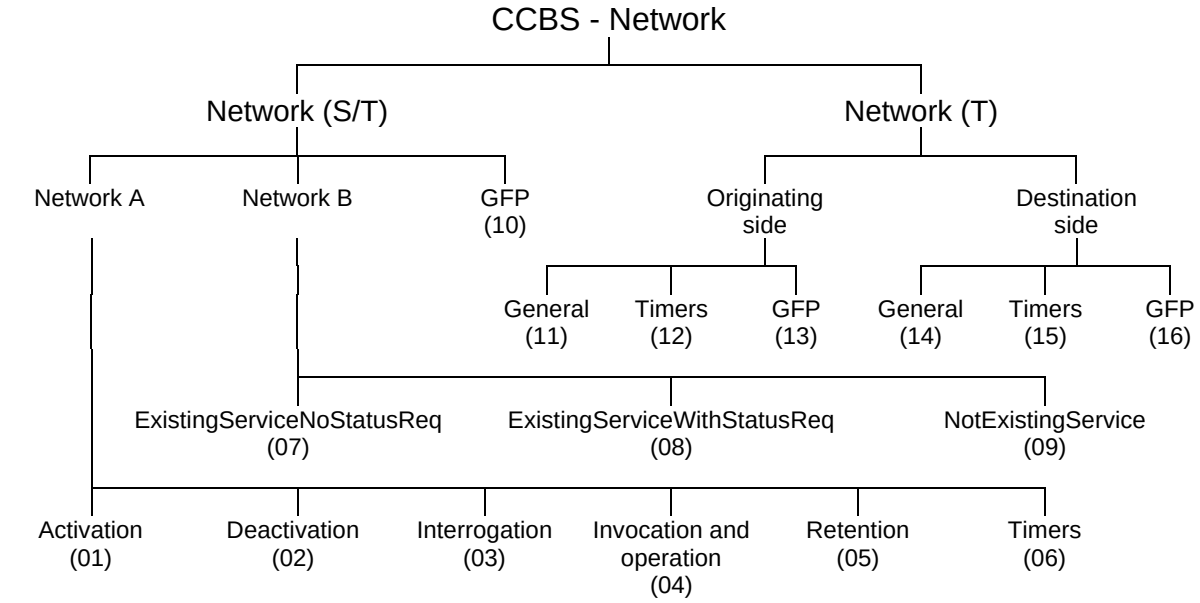
supplementary service: See ITU-T Recommendation I.210 [11], subclause 2.4.

4 Abbreviations

For the purposes of the present document, the following abbreviations apply:

ATM	Abstract Test Method
ATS	Abstract Test Suite
CCBS	Completion of Calls to Busy Subscriber
CR	Call Reference
CR1	normal (bearer related) CR
CR2	CR used for bearer independent transport mechanism
DSS1	Digital Subscriber Signalling System No. one
GFP	Generic Functional Protocol
ISDN	Integrated Services Digital Network
IUT	Implementation Under Test
N00	Null call state
N01	Call Initiated call state
N03	Outgoing Call Proceeding call state
N04	Call Delivered call state
N06	Call Present call state
N07	Call Received call state
N08	Connect Request call state
N09	Incoming Call Proceeding call state
N10	Active call state
N12	Disconnect Indication call state
N25	Overlap Receiving call state
N31	Bearer Independent Transport call state
PICS	Protocol Implementation Conformance Statement
PIXIT	Protocol Implementation eXtra Information for Testing
TP	Test Purpose
TSS	Test Suite Structure
UI	Unnumbered Information

5 Test Suite Structure (TSS)



NOTE: Numbers in brackets represent group numbers and are used in TP identifiers.

Figure 1: Test suite structure

6 Test Purposes (TP)

6.1 Introduction

For each test requirement a TP is defined.

6.1.1 TP naming convention

TPs are numbered, starting at 001, within each group. Groups are organized according to the TSS. Additional references are added to identify the actual supplementary service and whether it applies to the network or the user (see table 1).

Table 1: TP identifier naming convention scheme

Identifier: **<ss>_<iut><group>_<nnn>**				
<ss>	=	supplementary service:	e.g. "CCBS"	
<iut>	=	type of IUT:	U N	User Network
<group>	=	group	2 digit field representing group reference according to TSS	
<nnn>	=	sequential number	(001-999)	

6.1.2 Source of TP definition

The TPs are based on EN 300 359-1 [1].

6.1.3 TP structure

Each TP has been written in a manner which is consistent with all other TPs. The intention of this is to make the TPs more readable and checkable. A particular structure has been used and this is illustrated in table 2. This table should be read in conjunction with any TP, i.e. use a TP as an example to fully understand the table.

Table 2: Structure of a single TP

TP part	Text	Example
Header	<Identifier> *tab* <paragraph number in base ETS> *tab* <type of test> *tab* <condition> *CR.*	see table 1 subclause 0.0.0 valid, invalid, inopportune mandatory, optional, conditional
Stimulus	Ensure that the IUT in the <basic call state> and <supplementary service state> <trigger> *see below for message structure* *or* <goal>	 N10, N12, etc. CCBS Idle state receiving a XXXX message to request a ...
Reaction	<action> <conditions> *if the action is sending* *see below for message structure* <next action>, *etc.* and enters <supplementary service state> *and/or* and remains in the same state(s) *or* and enters state <state>	sends, saves, does, etc. using en-bloc sending, ...
Message structure	<message type> message containing a *a)* <info element> information element with *b)* a <field name> encoded as *or* including <coding of the field> and *back to a or b,*	SETUP, FACILITY, CONNECT, ... Bearer capability, Facility, ...
NOTE: Text in italics will not appear in TPs and text between <> is filled in for each TP and may differ from one TP to the next.		

6.1.4 Test strategy

As the base standard EN 300 359-1 [1] contains no explicit requirements for testing, the TPs were generated as a result of an analysis of the base standard and the PICS specification EN 300 359-2 [2]. The criteria applied include the following:

- only the requirements from the point of view of the T or coincident S and T reference point are considered;
- whether or not a test case can be built from the TP is not considered.

6.2 Network TPs for CCBS

6.2.1 Network (S/T)

NOTE 1: All FACILITY messages in TPs associated with clause 9, use the dummy call reference as specified in subclauses 8.3.2.2 and 8.3.2.4 of EN 300 196-1 [6] (bearer independent connectionless transport mechanism). Unless stated otherwise, FACILITY messages are sent/received using point-to-point data link (I frame) and the IUT is configured so that it "knows" that a point-to-point configuration exists at the user's access.

NOTE 2: Although the sending or receiving of a message using the dummy call reference is independent of any particular call state, in the following TPs call state N12 is used to show that the IUT has just begun clearing of a call and call state N00 is used to indicate that Layer 2 is active and capable of carrying bearer independent messages.

6.2.1.1 Network A

6.2.1.1.1 Activation

CCBS_N01_001 subclause 9.1.1 valid mandatory
Ensure that the IUT in the Disconnect Indication call state N12 and CCBS Idle state and Retention Active state for CCBS, on receipt of a FACILITY message containing a Facility information element with a CCBSRequest invoke component including the CallLinkageID,
sends a FACILITY message containing a Facility information element with a CCBSRequest return result component including the CCBSReference and recallMode and remains in call state N12.

CCBS_N01_002 subclause 9.1.2 inopportune optional
Ensure that the IUT in the Disconnect Indication call state N12 and CCBS Idle state and Retention Active state, on receipt of a FACILITY message containing a Facility information element with a CCBSRequest invoke component including the CallLinkageID from a user who has not subscribed to CCBS,
sends a FACILITY message containing a Facility information element with a CCBSRequest return error component indicating "notSubscribed" and remains in call state N12.

Selection: IUT provides Call Information Retention procedures even though CCBS not subscribed.

CCBS_N01_003 subclause 9.1.2 inopportune mandatory
Ensure that the IUT in the Disconnect Indication call state N12 and CCBS Idle state and Retention Active state for CCBS, on receipt of a FACILITY message containing a Facility information element with a CCBSRequest invoke component including an invalid CallLinkageID,
sends a FACILITY message containing a Facility information element with a CCBSRequest return error component indicating "invalidCallLinkageID" and remains in call state N12.

CCBS_N01_004 subclause 9.1.2 inopportune optional
Ensure that the IUT in the Disconnect Indication call state N12 and CCBS Idle state and Retention Active state, on receipt of a FACILITY message containing a Facility information element with a CCBSRequest invoke component including the CallLinkageID even though the attempted call failed for a reason other than the called user was busy,
sends a FACILITY message containing a Facility information element with a CCBSRequest return error component indicating "callFailureReasonNotBusy" and remains in call state N12.

Selection: IUT provides Call Information Retention procedures for service other than CCBS.

CCBS_N01_005 subclause 9.1.2 inopportune optional
Ensure that the IUT in the Disconnect Indication call state N12 and CCBS Idle state and Retention Active state, on receipt of a FACILITY message containing a Facility information element with a CCBSRequest invoke component including the CallLinkageID but user A's CCBS queue is full,
sends a FACILITY message containing a Facility information element with a CCBSRequest return error component indicating "outgoingCCBSQueueFull" and remains in call state N12.

Selection: IUT provides Call Information Retention procedures for service other than CCBS OR IUT provides Call Information Retention procedures for CCBS even when user A's CCBS queue is full.

CCBS_N01_006 subclause 9.1.2 inopportune mandatory
Ensure that the IUT in the Null call state N00 and CCBS Activated state, on receipt of a FACILITY message containing a Facility information element with a CCBSRequest invoke component including the CallLinkageID (but the served user has already activated the CCBS supplementary service for the call identified by the CallLinkageID),
sends a FACILITY message containing a Facility information element with a CCBSRequest return error component indicating "cCBSIsAlreadyActivated" and remains in call state N00.

CCBS_N01_007 subclause 9.1.2 inopportune optional
Ensure that the IUT in the Disconnect Indication call state N12 and CCBS Idle state and Retention Active state, on receipt of a FACILITY message containing a Facility information element with a CCBSRequest invoke component including the CallLinkageID but the served user has already activated CCBS supplementary service for an identical call (in Null call state N00 and CCBS Activated state),
sends a FACILITY message containing a Facility information element with a CCBSRequest return error component indicating "cCBSIsAlreadyActivated" and remains in call state N12.

Selection: IUT supports option to "Check for identical calls". PICS: MC 8.

CCBS_N01_008 subclause 9.1.2 inopportune mandatory
Ensure that the IUT in the Disconnect Indication call state N12 and CCBS Idle state and Retention Active state for CCBS, on receipt of a FACILITY message containing a Facility information element with a CCBSRequest invoke component including the CallLinkageID, but interactions between CCBS supplementary service and the call identified by the CallLinkageID are invalid,

sends a FACILITY message containing a Facility information element with a CCBSRequest return error component indicating "supplementaryServiceInteractionNotAllowed" and remains in call state N12.

CCBS_N01_009 subclause 9.1.2 inopportune optional
Ensure that the IUT in the Disconnect Indication call state N12 and CCBS Idle state and Retention Active state, on receipt of a FACILITY message containing a Facility information element with a CCBSRequest invoke component including the CallLinkageID, but CCBS is not available to the destination,

sends a FACILITY message containing a Facility information element with a CCBSRequest return error component indicating "longTermDenial" and remains in call state N12.

Selection: The IUT supports Call Information Retention procedure when "CCBS is not available to the destination".

CCBS_N01_010 subclause 9.1.2 inopportune mandatory
Ensure that the IUT in the Disconnect Indication call state N12 and CCBS Idle state and Retention Active state, on receipt of a FACILITY message containing a Facility information element with a CCBSRequest invoke component including the CallLinkageID, but CCBS is not available to the destination at this time,

sends a FACILITY message containing a Facility information element with a CCBSRequest return error component indicating "shortTermDenial" and remains in call state N12.

6.2.1.1.2 Deactivation

CCBS_N02_001 subclauses 9.2.1, 9.4.4.1 valid mandatory
Ensure that the IUT in the Null call state N00 and CCBS Activated state, on receipt of a FACILITY message containing a Facility information element with a CCBSDeactivate invoke component including the correct CCBSReference parameter,

sends a FACILITY message containing a Facility information element with a CCBSDeactivate return result component with cCBSEraseReason indicating "normal-unspecified" and a FACILITY message containing a Facility information element with a CCBSErase invoke component and enters the CCBS Idle state.

CCBS_N02_002 subclause 9.2.2 inopportune mandatory
Ensure that the IUT in the Null call state N00 and CCBS Activated state, on receipt of a FACILITY message containing a Facility information element with a CCBSDeactivate invoke component including an invalid CCBSReference,

sends a FACILITY message containing a Facility information element with a CCBSDeactivate return error component indicating "invalidCCBSReference" and remains in the same state.

CCBS_N02_003 subclause 9.2.2 inopportune mandatory
Ensure that the IUT in the Null call state N00 and CCBS Idle state, on receipt of a FACILITY message containing a Facility information element with a CCBSDeactivate invoke component when the user has not subscribed to the supplementary service,

sends a FACILITY message containing a Facility information element with a CCBSDeactivate return error component indicating "invalidCCBSReference".

6.2.1.1.3 Interrogation

CCBS_N03_001 subclause 9.3.1.1 valid mandatory
Ensure that the IUT in the Null call state N00 and CCBS Activated state, on receipt of a FACILITY message containing a Facility information element with a CCBSInterrogate invoke component without a CCBSReference parameter,

sends a FACILITY message containing a Facility information element with a CCBSInterrogate return result component including the correct value for the recallMode parameter, and in the CallDetails parameter a list of all currently active CCBS requests giving for each the CCBSReference, addressOfB, q931InfoElement and if available, the subAddressOfA.

CCBS_N03_002 subclause 9.3.1.1 valid mandatory
Ensure that the IUT in the Null call state N00 and CCBS Idle state, on receipt of a FACILITY message containing a Facility information element with a CCBSInterrogate invoke component without a CCBSReference parameter and no CCBS requests exist,
sends a FACILITY message containing a Facility information element with a CCBSInterrogate return result component including the correct value for the recallMode parameter, and no CallDetails parameter.

CCBS_N03_003 subclause 9.3.1.2 inopportune mandatory
Ensure that the IUT in the Null call state N00 and CCBS Idle state, on receipt of a FACILITY message containing a Facility information element with a CCBSInterrogate invoke component without a CCBSReference parameter but the user has not subscribed to CCBS,
sends a FACILITY message containing a Facility information element with a CCBSRequest return error component indicating "notSubscribed".

CCBS_N03_004 subclause 9.3.2.1 valid mandatory
Ensure that the IUT in the Null call state N00 and CCBS Activated state, on receipt of a FACILITY message containing a Facility information element with a CCBSInterrogate invoke component including a valid cCBSReference parameter,
sends a FACILITY message containing a Facility information element with a CCBSInterrogate return result component including the recallMode and in the callDetails parameter, addressOfB, q931InfoElement, cCBSReference and if available, the subAddressOfA.

CCBS_N03_005 subclause 9.3.2.2 inopportune mandatory
Ensure that the IUT in the Null call state N00 and CCBS Idle state, on receipt of a FACILITY message containing a Facility information element with a CCBSInterrogate invoke component including a CCBSReference parameter but the user has not subscribed to CCBS,
sends a FACILITY message containing a Facility information element with a CCBSRequest return error component indicating "notSubscribed".

CCBS_N03_006 subclause 9.3.2.2 inopportune mandatory
Ensure that the IUT in the Null call state N00 and CCBS Activated state, on receipt of a FACILITY message containing a Facility information element with a CCBSInterrogate invoke component including an invalid CCBSReference parameter,
sends a FACILITY message containing a Facility information element with a CCBSRequest return error component indicating "invalidCCBSReference".

6.2.1.1.4 Invocation and operation

CCBS_N04_001 subclause 9.4.1.1 valid mandatory
Ensure that the IUT in the Null call state N00 and CCBS Activated state, having checked that the user is neither busy nor CCBS busy, in order to indicate that it is prepared for establishment of the requested call,
sends a FACILITY message containing a Facility information element with a CCBSRemoteUserFree invoke component including the recallMode, cCBSReference, addressOfB and q931InfoElement.

CCBS_N04_002 subclause 9.4.1.1 valid mandatory
Ensure that the IUT in the Null call state N00 and CCBS Activated state, and a multipoint configuration exists, having checked that the user is neither busy nor CCBS busy, in order to indicate that it is prepared for establishment of the requested call,
sends a FACILITY message (UI frame) containing a Facility information element with a CCBSRemoteUserFree invoke component including the recallMode, cCBSReference, addressOfB and q931InfoElement.

CCBS_N04_003 subclause 9.4.1.2 valid mandatory
Ensure that the IUT in the Null call state N00 and CCBS Free state, (after sending a FACILITY message containing a Facility information element with a CCBSRemoteUserFree invoke component) on receipt of a FACILITY message with a CCBSRemoteUserFree reject component,
takes no action and remains in the same states.

CCBS_N04_004 subclause 9.4.2.1 valid optional
Ensure that the IUT in the Null call state N00 and CCBS Free state, if the specific recall option applies, on receipt of a SETUP message containing Bearer capability information element(s) from the original call and a Facility information element with a CCBSCall invoke component including the CCBSReference from the previously sent CCBSRemoteUserFree invoke component,
continues en-bloc basic call procedures using the retained call information and moves to call state N01.

Selection: Specific recall option supported. PICS: MC 7.2.

CCBS_N04_005 subclause 9.4.2.1 valid optional
Ensure that the IUT in the Null call state N00 and CCBS Free state, if the global recall option applies and a multipoint configuration exists, on receipt of a SETUP message containing Bearer capability information element(s) from the original call and a Facility information element with a CCBSCall invoke component including the CCBSReference from the previously sent CCBSRemoteUserFree invoke component,
continues basic call procedures and sends a FACILITY message (UI frame) containing a Facility information element with a CCBSStopAlerting invoke component including the CCBSReference and moves to call state N01.

Selection: Global recall option supported. PICS: MC 7.1.

CCBS_N04_006 subclause 9.4.2.2 inopportune mandatory
Ensure that the IUT in the Null call state N00 and CCBS Free state, on receipt of a SETUP message containing Bearer capability information element(s) from the original call and containing a Facility information element with a CCBSCall invoke component including an invalid CCBSReference value,
sends a RELEASE COMPLETE message containing a Facility information element with a CCBSCall return error component indicating "invalidCCBSReference" and moves to call state N00.

CCBS_N04_007 subclause 9.4.2.2 inopportune mandatory
Ensure that the IUT in the Null call state N00 and CCBS Activated state, on receipt of a SETUP message containing Bearer capability information element(s) from the original call and containing a Facility information element with a CCBSCall invoke component before having sent a cCBSRemoteUserFree invoke component for this user's CCBSReference,
sends a RELEASE COMPLETE message containing a Facility information element with a CCBSCall return error component indicating "notReadyForCall" and moves to call state N00.

CCBS_N04_008 subclause 9.4.2.2 inopportune optional
Ensure that the IUT in the Null call state N00 and CCBS Free state, on receipt of a SETUP message containing Bearer capability information element(s) from the original call and containing a Facility information element with a CCBSCall invoke component when no B-channels can be selected,
sends a RELEASE COMPLETE cause #34 or #44 and moves to call state N00.

Selection: IUT supports specific recall option. PICS: MC 7.2.

CCBS_N04_009 subclause 9.4.2.2 inopportune optional
Ensure that the IUT in the Null call state N00 and CCBS Free state, where a multipoint configuration exists and the global recall option applies, on receipt of a SETUP message containing Bearer capability information element(s) from the original call and containing a Facility information element with a CCBSCall invoke component when no B-channels can be selected,
sends a FACILITY message (UI frame) containing a Facility information element with a CCBSStopAlerting invoke component including the same CCBSReference value and a RELEASE COMPLETE cause #34 or #44 and moves to call state N00.

Selection: Global recall option supported. PICS: MC 7.1.

CCBS_N04_010 subclause 9.4.2.2 inopportune optional
Ensure that the IUT in the Null call state N00 and CCBS Free state, where a multipoint configuration exists and the global recall option applies, on receipt of more than one SETUP message containing Bearer capability information element(s) from the original call and containing a Facility information element with a CCBSCall invoke component,
continues basic call procedures for the first SETUP message and sends a RELEASE COMPLETE message containing a Facility information element with a CCBSCall return error component indicating "alreadyAccepted" in response to the other SETUP messages and moves to call state N01.

Selection: Global recall option supported. PICS: MC 7.1.

CCBS_N04_011 subclauses 9.4.3.1, 9.4.4.1 valid optional
Ensure that the IUT in the Outgoing call proceeding call state N03 and CCBS Call Init state, to indicate that user B has responded to the call with an ALERTING message,
sends an ALERTING message followed by a FACILITY message containing a Facility information element with a cCBSErase invoke indicating cCBSEraseReason "normal-unspecified" and enters the call state N04.

CCBS_N04_012 subclauses 9.4.3.1, 9.4.4.1 valid optional
Ensure that the IUT in the Outgoing call proceeding call state N03 and CCBS Call Init state, to indicate that user B has responded to the call with a CONNECT message,
sends a CONNECT message followed by a FACILITY message containing a Facility information element with a cCBSErase invoke indicating cCBSEraseReason "normal-unspecified" and enters the call state N10.

CCBS_N04_013 subclauses 9.4.3.2, 9.4.4.1 valid optional
Ensure that the IUT in the Outgoing Call Proceeding call state N03 and CCBS Call Init state, if it is not able to establish the call because the called user is busy again,
sends a DISCONNECT message not containing a Facility information element with a cCBSErase invoke component and enters the call state N12 or N00.

Selection: "CCBS request retention" option supported. PICS: MC 6.

CCBS_N04_014 subclauses 9.4.3.2, 9.4.4.1 valid optional
Ensure that the IUT in the Outgoing Call Proceeding call state N03 and CCBS Call Init state, if it is not able to establish the call because the called user is busy again,
sends a DISCONNECT message containing a Facility information element with a CallInfoRetain invoke component including a CallLinkageID; and
sends a FACILITY message containing a Facility information element with a CCBSErase invoke component including CCBSEraseReason encoded as "basic-call-failed"; and
enters call state N12.

Selection: "CCBS request retention" option NOT supported. PICS: NOT MC 6.

CCBS_N04_015 subclauses 9.4.3.2, 9.4.4.1 valid optional
Ensure that the IUT in the Outgoing Call Proceeding call state N03 and CCBS Call Init state, where a multipoint configuration exists, if it is not able to establish the call because the called user is busy again,
sends a DISCONNECT message containing a Facility information element with a CallInfoRetain invoke component including a CallLinkageID,; and
sends a FACILITY message (UI frame) containing a Facility information element with a CCBSErase invoke component including CCBSEraseReason encoded as "basic-call-failed"; and
enters call state N12.

Selection: "CCBS request retention" option NOT supported. PICS: NOT MC 6.

CCBS_N04_016 subclauses 9.4.3.2, 9.4.4.1 valid optional
Ensure that the IUT in the Outgoing Call Proceeding call state N03 and CCBS Call Init state, if it is not able to establish the call for any reason other than the called user is busy,
sends a DISCONNECT message; and
sends a FACILITY message containing a Facility information element with a CCBSErase invoke component including CCBSEraseReason encoded as "basic-call-failed"; and
enters call state N12.

Selection: "CCBS request retention" option supported. PICS: MC 6.

CCBS_N04_017 subclauses 9.4.3.2, 9.4.4.1 valid optional
Ensure that the IUT in the Outgoing Call Proceeding call state N03 and CCBS Call Init state, where a multipoint configuration exists, if it is not able to establish the call for any reason other than the called user is busy,
sends a DISCONNECT message; and
sends a FACILITY message (UI frame) containing a Facility information element with a CCBSErase invoke component including CCBSEraseReason encoded as "basic-call-failed"; and
enters call state N12.

Selection: "CCBS request retention" option supported. PICS: MC 6.

NOTE 1: The above two TPs are now repeated but with the "CCBS request retention" option NOT supported. This is to demonstrate that the deactivation of the CCBS supplementary service under these circumstances is independent of this option.

CCBS_N04_018 subclauses 9.4.3.2, 9.4.4.1 valid optional
Ensure that the IUT in the Outgoing Call Proceeding call state N03 and CCBS Call Init state, if it is not able to establish the call for any reason other than the called user is busy,
sends a DISCONNECT message; and
sends a FACILITY message containing a Facility information element with a CCBSErase invoke component including CCBSEraseReason encoded as "basic-call-failed"; and
enters call state N12.

Selection: "CCBS request retention" option NOT supported. PICS: NOT MC 6.

CCBS_N04_019 subclauses 9.4.3.2, 9.4.4.1 valid optional
Ensure that the IUT in the Outgoing Call Proceeding call state N03 and CCBS Call Init state, where a multipoint configuration exists, if it is not able to establish the call for any reason other than the called user is busy,
sends a DISCONNECT message; and
sends a FACILITY message (UI frame) containing a Facility information element with a CCBSErase invoke component including CCBSEraseReason encoded as "basic-call-failed"; and
enters call state N12.

Selection: "CCBS request retention" option NOT supported. PICS: NOT MC 6.

CCBS_N04_020 subclauses 9.4.3.2, 9.4.4.1 inopportune mandatory
Ensure that the IUT in the Outgoing Call Proceeding call state N03 and CCBS Call Init state, on receipt of a DISCONNECT message from the served user (before the IUT has sent an ALERTING or CONNECT message),
sends a RELEASE message; and
sends a FACILITY message containing a Facility information element with a CCBSErase invoke component indicating "basic-call-failed"; and
enters call state N19.

CCBS_N04_021 subclauses 9.4.3.2, 9.4.4.1 inopportune mandatory
Ensure that the IUT in the Outgoing Call Proceeding call state N03 and CCBS Call Init state, where a multipoint configuration exists, on receipt of a DISCONNECT message from the served user (before the IUT has sent an ALERTING or CONNECT message),
sends a RELEASE message; and
sends a FACILITY message (UI frame) containing a Facility information element with a CCBSErase invoke component indicating "basic-call-failed"; and
enters call state N19.

CCBS_N04_022 subclauses 9.4.3.2, 9.4.4.1 inopportune mandatory
Ensure that the IUT in the Call Delivered call state N04 and CCBS Call Init state, on receipt a FACILITY message containing a Facility information element with a CCBSDeactivate invoke component,
continues with normal call handling; and
sends a FACILITY message containing a Facility information element with a CCBSDeactivate return result component; and
sends a FACILITY message containing a Facility information element with a CCBSErase invoke component indicating "normal-unspecified"; and
remains in the call state N04.

CCBS_N04_023 subclauses 9.4.3.2, 9.4.4.1 inopportune mandatory
Ensure that the IUT in the Call Delivered call state N04 and CCBS Call Init state, where a multipoint configuration exists, on receipt a FACILITY message containing a Facility information element with a CCBSDeactivate invoke component,
continues with normal call handling; and
sends a FACILITY message (I frame) containing a Facility information element with a CCBSDeactivate return result component; and
sends a FACILITY message (UI frame) containing a Facility information element with a CCBSErase invoke component indicating "normal-unspecified"; and
remains in the call state N04.

CCBS_N04_024 subclause 9.4.5.1 valid mandatory
Ensure that the IUT in the Null call state N00 and CCBS Activated state, and having determined that the served user is either busy or CCBS busy,
sends a FACILITY message containing a Facility information element with a CCBSBFree invoke component including the recallMode, CCBSReference, addressOfB and q931InfoElement, and remains in call state N00.

NOTE 2: In this TP, the IUT need to be informed (internal network indication) that user B is not busy.

CCBS_N04_025 subclause 9.4.5.1 valid mandatory
Ensure that the IUT in the Null call state N00 and CCBS Activated state, where a multipoint configuration exists, and having determined that the served user is either busy or CCBS busy,
sends a FACILITY message (UI frame) containing a Facility information element with a CCBSBFree invoke component including the recallMode, CCBSReference, addressOfB and q931InfoElement and remains in call state N00.

NOTE 3: In this TP the IUT needs to be informed (internal network indication) that user B is not busy.

CCBS_N04_026 subclause 9.4.5.2 valid mandatory
Ensure that the IUT in the Null call state N00 and CCBS Suspended state, having sent a CCBSFree invoke component, on receipt of a FACILITY message containing a Facility information element with a CCBSBFree reject component,
takes no action and remains in call state N00.

CCBS_N04_027 subclause 9.4.6.1 valid mandatory
Ensure that the IUT in the Null call state N00 and CCBS Activated state, to determine if the served user is not busy,
sends a FACILITY message containing a Facility information element with a CCBSStatusRequest invoke component including the recallMode, CCBSReference and q931InfoElement and enters CCBS Check A state and remains in call state N00.

CCBS_N04_028 subclause 9.4.6.1 valid mandatory
Ensure that the IUT in the Null call state N00 and CCBS Activated state, where a multipoint configuration exists, to determine if the served user is not busy,
sends a FACILITY message (UI frame) containing a Facility information element with a CCBSStatusRequest invoke component including the recallMode, CCBSReference and q931InfoElement and remains in call state N00.

CCBS_N04_029 subclause 9.4.6.1 valid mandatory
Ensure that the IUT in the Null call state N00 and CCBS Check A state, on receipt of a CCBSStatusRequest return result component indicating "busy",
takes no protocol actions.

CCBS_N04_030 subclause 9.4.6.1 valid mandatory
Ensure that the IUT in the Null call state N00 and CCBS Check A state, on receipt of a CCBSStatusRequest return result component indicating "free",
takes no protocol actions.

CCBS_N04_031 subclause 9.4.6.1 valid mandatory
Ensure that the IUT in the Null call state N00 and CCBS Check A state, where a multipoint configuration exists, on receipt of multiple CCBSStatusRequest return result components,
takes no protocol actions.

CCBS_N04_032 subclause 9.4.6.2 valid mandatory
Ensure that the IUT in the Null call state N00 and CCBS Check A state, having sent a FACILITY message containing a Facility information element with a CCBSStatusRequest invoke component, on receipt of a FACILITY message containing a Facility information element with a CCBSStatusRequest reject component,
takes no protocol action.

6.2.1.1.5 Retention

CCBS_N05_001 subclause 9.6.1 valid mandatory
Ensure that the IUT in the Outgoing Call Proceeding call state N03 and Retention Idle state, to provide the call information retention procedure,
sends a DISCONNECT message containing a Facility information element with a CallInfoRetain invoke component including a CallLinkageID and enters state N12 and Retention Active state.

CCBS_N05_002 subclause 9.6.1 valid conditional
Ensure that the IUT in the Null call state N00 and Retention Active state and CCBS Activated state, having released the call information on operation of the CCBS supplementary service, and it is the case that no other supplementary service needs the call information,
sends a FACILITY message containing a Facility information element with an EraseCallLinkageID invoke component including the CallLinkageID.

CCBS_N05_003 subclause 9.6.1 valid conditional
Ensure that the IUT in the Null call state N00 and Retention Active state and CCBS Activated state, where a multipoint configuration exists, having released the call information on operation of the CCBS supplementary service, and it is the case that no other supplementary service needs the call information,
sends a FACILITY message (UI frame) containing a Facility information element with an EraseCallLinkageID invoke component including the CallLinkageID.

CCBS_N05_004 subclause 9.6.2 valid conditional
Ensure that the IUT in the Null call state N00 and CCBS Activated state, having sent a FACILITY message containing a Facility information element with an EraseCallLinkageID invoke component including the CallLinkageID to the served user, on receipt of a FACILITY message containing a Facility information element with an EraseCallLinkageID reject component,
sends no frame and remains in the call state N00.

6.2.1.1.6 Timers

CCBS_N06_001 subclause 9.4.1.2 timer mandatory
Ensure that the IUT in the Null call state N00 and CCBS Free state and if timer T-CCBS3 expires,
sends a FACILITY message containing a Facility information element with the CCBSErase invoke component including the CCBSEraseReason coded as "t-CCBS3-timeout" and enters CCBS Idle state.

CCBS_N06_002 subclause 9.4.1.2 timer mandatory
Ensure that the IUT in the Null call state N00 and CCBS Free state and a multipoint configuration exists and if timer T-CCBS3 expires,
sends a FACILITY message (UI frame) containing a Facility information element with the CCBSErase invoke component including the CCBSEraseReason coded as "t-CCBS3-timeout".

CCBS_N06_003 subclauses 9.4.3.2, 9.4.1.2 timer mandatory
Ensure that the IUT in the Null call state N00 and CCBS Activated state, on expiry of timer T-CCBS2,
sends a FACILITY message containing a Facility information element with the CCBSErase invoke component including the CCBSEraseReason coded as "t-CCBS2-timeout".

CCBS_N06_004 subclauses 9.4.3.2, 9.4.1.2 timer mandatory
Ensure that the IUT in the Null call state N00 and CCBS Activated state, where a multipoint configuration exists, on expiry of timer T-CCBS2,
sends a FACILITY message (UI frame) containing a Facility information element with the CCBSErase invoke component including the CCBSEraseReason coded as "t-CCBS2-timeout".

CCBS_N06_005 subclauses 9.4.6.2, 9.4.4 timer mandatory
Ensure that the IUT in the Null call state N00 and CCBS Check A state, having sent a FACILITY message containing a Facility information element with a CCBSStatusRequest invoke component, if the timer T-CCBS1 expires and no FACILITY message containing a Facility information element with a CCBSStatusRequest return result component has been received,
sends a FACILITY message containing a Facility information element with a CCBSErase invoke component indicating "normal-unspecified".

CCBS_N06_006 subclauses 9.4.6.2, 9.4.4 timer mandatory
Ensure that the IUT in the Null call state N00 and CCBS Check A state, where a multipoint configuration exists, having sent a FACILITY message containing a Facility information element with a CCBSStatusRequest invoke component, if the timer T-CCBS1 expires and no FACILITY message containing a Facility information element with a CCBSStatusRequest return result component has been received,
sends a FACILITY message (UI frame) containing a Facility information element with a CCBSErase invoke component indicating "normal-unspecified".

CCBS_N06_007 subclause 9.6.1 timer conditional
Ensure that the IUT in N12 and CCBS Idle state, on expiry of T-RETENTION,
sends a FACILITY message containing a Facility information element with an EraseCallLinkageID invoke component containing the callLinkageID parameter.

Selection: The IUT provides the call information retention procedure. PICS: MC 5.

CCBS_N06_008 subclause 9.6.1 timer conditional
Ensure that the IUT in N12 and CCBS Idle state, where a multipoint configuration exists, on expiry of T-RETENTION,
sends a FACILITY message (UI frame) containing a Facility information element with an EraseCallLinkageID invoke component containing the callLinkageID parameter.

Selection: The IUT provides the call information retention procedure. PICS: MC 5.

NOTE: Timers T-ACTIVATE, T-DEACTIVATE, T-INTERROGATE are user A timers and so are not included in the present document covering the network side test specification.

6.2.1.2 Network B

NOTE: These procedures apply to the interface between network B and user B (non-served user). The subscription option "status request procedures for existing services" (value = supported/not supported) is a user B option. The network should support both options, i.e. be capable of being configured for both options for one particular user interface.

6.2.1.2.1 ExistingServiceNoStatusReq

CCBS_N07_001 subclause 9.5.4.1 valid mandatory
Ensure that the IUT in N00, when there is a free B-channel and the subscription parameter "status request procedures for existing services" is set to "not supported", and the service is an existing service, in order to determine whether user B is free,
starts timer T-CCBS4 and reserves a B-channel, sends no message and remains in the same state.

CCBS_N07_002 subclause 9.5.4.1 valid mandatory
Ensure that the IUT in N00, having reserved a B-channel and a free B-channel exists, and the subscription parameter "status request procedures for existing services" is set to "not supported", and the service is an existing service, on expiry of timer T-CCBS4,
continues to reserve a B-channel, informs network A that user B is free, sends no message and remains in the same state.

6.2.1.2.2 ExistingServiceWithStatusReq

CCBS_N08_001 subclause 9.5.4.1 valid mandatory
Ensure that the IUT in state N00, when there is a free B-channel and the subscription parameter "status request procedures for existing services" is set to "supported", and the service is an existing service, in order to determine whether user B is free,
reserves a B-channel and sends a FACILITY message using the dummy call reference containing a Facility information element with a StatusRequest invoke component.

CCBS_N08_002 subclause 9.5.4.1 valid mandatory
Ensure that the IUT in N00 and the Waiting Status state, having reserved a B-channel, and the subscription parameter "status request procedures for existing services" is set to "supported", and the service is an existing service, on receipt of a FACILITY message using the dummy call reference containing a Facility information element with a StatusRequest return result component indicating "compatibleAndFree",
starts timer T-CCBS4, sends no message and remains in the same state.

CCBS_N08_003 subclause 9.5.4.1 valid mandatory
Ensure that the IUT in N00 and the Waiting Status state, having reserved a B-channel, and the subscription parameter "status request procedures for existing services" is set to "supported", and the service is an existing service, on receipt of a FACILITY message using the dummy call reference containing a Facility information element with a StatusRequest return result component indicating "compatibleAndBusy",
cancels the B-channel reservation, sends no message and remains in the same state.

CCBS_N08_004 subclause 9.5.4.1 valid mandatory
Ensure that the IUT in N00, having reserved a B-channel and there being a free B-channel, and the subscription parameter "status request procedures for existing services" is set to "supported", and the service is an existing service, on expiry of timer T-CCBS4,
continues to reserve a B-channel and sends a FACILITY message using the dummy call reference containing a Facility information element with a StatusRequest invoke component.

CCBS_N08_005 subclause 9.5.4.1 valid mandatory
Ensure that the IUT in N00 and the Waiting Status state, having reserved a B-channel, and timer T-CCBS4 having expired and the subscription parameter "status request procedures for existing services" is set to "supported", and the service is an existing service, on receipt of a FACILITY message using the dummy call reference containing a Facility information element with a StatusRequest return result component indicating "compatibleAndFree",
sends no message and remains in the same state.

CCBS_N08_006 subclause 9.5.4.1 valid mandatory
Ensure that the IUT in N00 and the Waiting Status state, having reserved a B-channel, and timer T-CCBS4 having expired and the subscription parameter "status request procedures for existing services" is set to "supported", and the service is an existing service, on receipt of a FACILITY message using the dummy call reference containing a Facility information element with a StatusRequest return result component indicating "compatibleAndBusy",
cancels any B-channel reservation, sends no message and remains in the same state.

CCBS_N08_007 subclause 9.5.4.2 valid mandatory
Ensure that the IUT in N00 and the Waiting Status state, having reserved a B-channel, and the subscription parameter "status request procedures for existing services" is set to "supported", and the service is an existing service, on receipt of a FACILITY message using the dummy call reference containing a Facility information element with a StatusRequest return result component indicating "incompatible",
cancels the B-channel reservation and deactivates the CCBS supplementary service.

CCBS_N08_008 subclause 9.5.4.2 valid mandatory
Ensure that the IUT in N00, having reserved a B-channel and no free B-channel is available, and the subscription parameter "status request procedures for existing services" is set to "supported", and the service is an existing service, on expiry of timer T-CCBS4,
cancels any B-channel reservation and waits for a B-channel to become free, sends no message and remains in the same state.

6.2.1.2.3 NotExistingService

CCBS_N09_001 subclause 9.5.4.1 valid mandatory
Ensure that the IUT in state N00, when there is a free B-channel and the service is NOT an existing service, in order to determine whether user B is free,
reserves a B-channel and sends a FACILITY message using the dummy call reference containing a Facility information element with a StatusRequest invoke component.

CCBS_N09_002 subclause 9.5.4.1 valid mandatory
Ensure that the IUT in N00 and the Waiting Status state, having reserved a B-channel, and the service is NOT an existing service, on receipt of a FACILITY message using the dummy call reference containing a Facility information element with a StatusRequest return result component indicating "compatibleAndFree",
starts timer T-CCBS4, sends no message and remains in the same state.

CCBS_N09_003 subclause 9.5.4.1 valid mandatory
Ensure that the IUT in N00 and the Waiting Status state, having reserved a B-channel, and the service is NOT an existing service, on receipt of a FACILITY message using the dummy call reference containing a Facility information element with a StatusRequest return result component indicating "compatibleAndBusy",
cancels the B-channel reservation, sends no message and remains in the same state.

CCBS_N09_004 subclause 9.5.4.1 valid mandatory
Ensure that the IUT in N00, having reserved a B-channel and there being a free B-channel, and the service is NOT an existing service, on expiry of timer T-CCBS4,
continues to reserve a B-channel and sends a FACILITY message using the dummy call reference containing a Facility information element with a StatusRequest invoke component.

CCBS_N09_005 subclause 9.5.4.1 valid mandatory
Ensure that the IUT in N00 and the Waiting Status state, having reserved a B-channel, and timer T-CCBS4 having expired and the service is NOT an existing service, on receipt of a FACILITY message using the dummy call reference containing a Facility information element with a StatusRequest return result component indicating "compatibleAndFree",
sends no message and remains in the same state.

CCBS_N09_006 subclause 9.5.4.1 valid mandatory
Ensure that the IUT in N00 and the Waiting Status state, having reserved a B-channel, and timer T-CCBS4 having expired and the service is NOT an existing service, on receipt of a FACILITY message using the dummy call reference containing a Facility information element with a StatusRequest return result component indicating "compatibleAndBusy",
cancels any B-channel reservation, sends no message and remains in the same state.

CCBS_N09_007 subclause 9.5.4.2 valid mandatory
Ensure that the IUT in N00 and the Waiting Status state, having reserved a B-channel, and the service is NOT an existing service, on receipt of a FACILITY message using the dummy call reference containing a Facility information element with a StatusRequest return result component indicating "incompatible",
cancels the B-channel reservation and deactivates the CCBS supplementary service, sends no message and remains in the same state.

CCBS_N09_008 subclause 9.5.4.2 valid mandatory
Ensure that the IUT in N00, having reserved a B-channel and no free B-channel is available, and the service is NOT an existing service, on expiry of timer T-CCBS4,
cancels any B-channel reservation and waits for a B-channel to become free, sends no message and remains in the same state.

6.2.1.3 GFP

CCBS_N10_001 clause 9, [6] subclause 8.3.2.2.2 invalid mandatory
Ensure that the IUT in call state N00 and in the CCBS Activated state receiving a FACILITY message containing a Facility information element with an invalid protocol profile,
ignores the message.

CCBS_N10_002 clause 9, [6] subclause 8.3.2.2.2 invalid mandatory
Ensure that the IUT in call state N00 and in the CCBS Activated state receiving FACILITY message without a Facility information element,
ignores the message.

CCBS_N10_003 clause 9, [6] subclause 8.3.2.2.2 invalid mandatory
Ensure that the IUT in call state N00 and in the CCBS Activated state receiving a message other than FACILITY with a dummy call reference and this message does not apply to some other application of the dummy call reference,
ignores the message.

CCBS_N10_004 clause 9, [6] subclause 8.4.2 invalid mandatory
Ensure that the IUT in call state N00 and in the CCBS Activated state on receipt of a FACILITY message containing a Facility information element with a CCBSInterrogate invoke component including a CCBSReference parameter of incorrect type,
ignores the optional CCBSReference parameter and does not reject the component with problem code of "mistyped argument".

6.2.2 Network (T)

NOTE: The private network procedures use the bearer independent connection-oriented transport mechanism as well as the bearer related transport mechanism. Different Call References (CRs) are used to differentiate between the two mechanisms. In the following TPs, these are identified by CR followed by a number:

CR1 = normal (bearer related) call reference;

CR2 = call reference used for bearer independent transport mechanism.

The values of CR1 and CR2 may vary from one TP to another, but when both are used in the same TP their values are distinct. CR1 and CR2 could exist at different exchanges.

6.2.2.1 Originating side

6.2.2.1.1 General

CCBS_N11_001 subclause 10.1.1.1 valid mandatory
Ensure that the IUT in the CCBS Idle state, with CR1 in call state N03, to indicate that a busy destination has been encountered,
sends a DISCONNECT message with CR1 and cause #17 or #34, containing a Facility information element with a CCBS-T-Available invoke component and moves to the call state N12.

CCBS_N11_002 subclause 10.1.2.1 valid optional
Ensure that the IUT in the CCBS Idle state, with CR2 in call state N00, on receipt of a REGISTER message with CR2 containing a Facility information element with a CCBS-T-Request invoke component including the retentionSupported parameter set to TRUE,
sends a FACILITY message with CR2 containing a Facility information element with a CCBS-T-Request return result component including the parameter retentionSupported set to TRUE and enters the Bearer Independent Transport state (N31) for CR2.

Selection: The IUT supports the CCBS request retention option. PICS: MC 6.

CCBS_N11_003 subclause 10.1.2.1 valid optional
Ensure that the IUT in the CCBS Idle state, with CR2 in call state N00, on receipt of a REGISTER message with CR2 containing a Facility information element with a CCBS-T-Request invoke component including the retentionSupported parameter set to TRUE,
sends a FACILITY message with CR2 containing a Facility information element with a CCBS-T-Request return result component including the parameter retentionSupported set to FALSE and enters the Bearer Independent Transport state (N31) for CR2.

Selection: The IUT does NOT support the CCBS request retention option. PICS: NOT MC 6.

CCBS_N11_004 subclause 10.1.2.2 valid mandatory
Ensure that the IUT in the CCBS Idle state, with CR2 in call state N00, on receipt of REGISTER message with CR2 containing a Facility information element with a (CCBS-related invoke component different from CCBS-T-Request invoke component) CCBS-T-Call invoke component,
sends a RELEASE with CR2 and cause #29 and enters the Release Request state (N19).

CCBS_N11_005 subclause 10.1.2.2 valid mandatory
Ensure that the IUT in the CCBS Idle state, with CR2 in call state N00, on receipt of REGISTER message with CR2 containing a Facility information element with a (CCBS-related invoke component different from CCBS-T-Request invoke component) CCBS-T-Suspend invoke component,
sends a RELEASE with CR2 and cause #29 and enters the Release Request state (N19).

CCBS_N11_006 subclause 10.1.2.2 valid mandatory
Ensure that the IUT in the CCBS Idle state, with CR2 in call state N00, on receipt of REGISTER message with CR2 containing a Facility information element with a (CCBS-related invoke component different from CCBS-T-Request invoke component) CCBS-T-Resume invoke component,
sends a RELEASE with CR2 and cause #29 and enters the Release Request state (N19).

CCBS_N11_007 subclause 10.1.2.2 valid mandatory
Ensure that the IUT in the CCBS Idle state, with CR2 in call state N00, on receipt of REGISTER message with CR2 containing a Facility information element with a (CCBS-related invoke component different from CCBS-T-Request invoke component) CCBS-T-RemoteUserFree invoke component,
sends a RELEASE with CR2 and cause #29 and enters the Release Request state (N19).

CCBS_N11_008 subclause 10.1.2.2 valid mandatory
Ensure that the IUT in the CCBS Idle state, with CR2 in call state N00, on receipt of REGISTER message with CR2 containing a Facility information element with a (CCBS-related invoke component different from CCBS-T-Request invoke component) CCBS-T-Available invoke component,
sends a RELEASE with CR2 and cause #29 and enters the Release Request state (N19).

CCBS_N11_009 subclause 10.1.2.2 inopportune mandatory
Ensure that the IUT in the CCBS Idle state, with CR2 in call state N00, on receipt of REGISTER message with CR2 containing a Facility information element with a CCBS-T-Request invoke component but the supplementary service CCBS is not subscribed to,
sends a FACILITY message with CR2 containing a Facility information element with a CCBS-T-Request return error component indicating "notSubscribed" and then sends a RELEASE message with cause #31 with CR2 to clear the signalling connection and enters state N19 for CR2;
or
sends a RELEASE message with CR2 and cause #31 containing a Facility information element with a CCBS-T-Request return error component indicating "notSubscribed" and enters state N19 for CR2.

CCBS_N11_010 subclause 10.1.2.2 inopportune mandatory
Ensure that the IUT in the CCBS Idle state, with CR2 in call state N00, on receipt of REGISTER message with CR2 containing a Facility information element with a CCBS-T-Request invoke component but the supplementary service CCBS is not available to the destination,
sends a FACILITY message with CR2 containing a Facility information element with a CCBS-T-Request return error component indicating "longTermDenial" and then sends a RELEASE message with cause #31 with CR2 to clear the signalling connection and enters state N19 for CR2;
or
sends a RELEASE message with CR2 and cause #31 containing a Facility information element with a CCBS-T-Request return error component indicating "longTermDenial" and enters state N19 for CR2.

CCBS_N11_011 subclause 10.1.2.2 inopportune mandatory
Ensure that the IUT in the CCBS Idle state, with CR2 in call state N00, on receipt of REGISTER message with CR2 containing a Facility information element with a CCBS-T-Request invoke component but the supplementary service CCBS is not available to the destination at this time,
sends a FACILITY message with CR2 containing a Facility information element with a CCBS-T-Request return error component indicating "shortTermDenial" and then sends a RELEASE message with cause #31 with CR2 to clear the signalling connection and enters state N19 for CR2;
or
sends a RELEASE message with CR2 and cause #31 containing a Facility information element with a CCBS-T-Request return error component indicating "shortTermDenial" and enters state N19 for CR2.

CCBS_N11_012 subclause 10.1.3.1 valid mandatory
Ensure that the IUT in the CCBS Free state, with CR2 in call state N31, to indicate that the destination has become not busy and that the IUT is ready to accept a call,
sends a FACILITY message with CR2 containing a Facility information element with a CCBS-T-RemoteUserFree invoke component.

CCBS_N11_013 subclause 10.1.3.2 valid mandatory
Ensure that the IUT in the CCBS Free state, with CR2 in call state N31, having sent a FACILITY message with CR2 containing a Facility information element with a CCBS-T-RemoteUserFree invoke component, on receipt of a FACILITY message with CR2 containing a Facility information element with a CCBS-T-RemoteUserFree reject component,
sends a RELEASE with CR2 and cause #31 and enters state N19 for CR2.

CCBS_N11_014 subclause 10.1.4.1, 2nd paragraph valid mandatory
Ensure that the IUT in the CCBS Free state, with CR2 in call state N31, on receipt of a FACILITY message with CR2 containing a Facility information element with a CCBS-T-Suspend invoke component,
takes no protocol actions.

CCBS_N11_015 subclause 10.1.5.1, 2nd paragraph valid mandatory
Ensure that the IUT in the CCBS Free state, with CR2 in call state N31, on receipt of a FACILITY message with CR2 containing a Facility information element with a CCBS-T-Resume invoke component,
takes no protocol actions.

CCBS_N11_016 subclause 10.1.6.1, 2nd paragraph valid mandatory
Ensure that the IUT in the CCBS Free state, with CR1 in call state N00 and with CR2 in call state N31, on receipt of a SETUP with CR1 using the call establishment information used in the original call attempt and including a Facility information element with a CCBS-T-Call invoke component,
sends a SETUP ACKNOWLEDGE or CALL PROCEEDING with CR1 and moves to the call state N02 or N03.

CCBS_N11_017 subclauses 10.1.6.1, [6] 8.3.2.1.3.1 valid mandatory
Ensure that the IUT in the CCBS Free state, with CR1 in call state N03 and with CR2 in call state N31, having sent an ALERTING or CONNECT message with CR1, to clear the signalling connection,
sends a RELEASE with CR2 and enters state N19 for CR2.

CCBS_N11_018 subclause 10.1.6.2, 1st paragraph inopportune optional
Ensure that the IUT in the CCBS Init state, with CR1 in call state N03 and with CR2 in call state N31, where the retention option is being used, to indicate that the called user is busy again,
sends a DISCONNECT message with CR1 containing a T-CCBS-Available invoke component and moves to call state N12.

Selection: CCBS request retention option supported. PICS: MC 6.

CCBS_N11_019 subclause 10.1.6.2, 3rd paragraph inopportune optional
Ensure that the IUT in the CCBS Init state, with CR1 in call state N03 and with CR2 in call state N31, where the retention option is not being used, to indicate that the called user is busy again and to clear the signalling connection,
sends a DISCONNECT message with CR1 containing a T-CCBS-Available invoke component to clear the attempted call and a RELEASE message with CR2 to clear the signalling connection and moves to call state N12 for CR1 and call state N19 for CR2.

Selection: CCBS request retention option NOT supported. PICS: NOT MC 6.

CCBS_N11_020 subclause 10.1.6.2, 4th paragraph inopportune mandatory
Ensure that the IUT in the CCBS Init state, with CR1 in call state N03 and with CR2 in call state N31, to indicate that the call failed at the destination side due to any reason other than the user at that side is busy,
sends a DISCONNECT message with CR1 to clear the attempted call and a RELEASE message with CR2 to clear the signalling connection and moves to call state N12 for CR1 and call state N19 for CR2.

CCBS_N11_021 subclause 10.1.6.2 valid mandatory
Ensure that the IUT in the CCBS Init state, with CR1 in call state N01 and with CR2 in call state N31, but the call fails before reaching the destination,
sends a DISCONNECT message with CR1 to clear the attempted call and moves to call state N12 for CR1.

CCBS_N11_022 subclause 10.1.7.1 valid mandatory
Ensure that the IUT in the CCBS Activated state, with CR1 in call state N03 and with CR2 in call state N31, in order to deactivate the CCBS request,
sends a RELEASE message with CR2 and with cause #31 and moves to call state N19 for CR2.

6.2.2.1.2 Timers

CCBS_N12_001 subclause 10.1.6.2 timer mandatory
Ensure that the IUT in the CCBS Free state, with CR2 in call state N31 and if timer T-CCBS6 expires,
sends a RELEASE message with CR2 and moves to call state N19 for CR2.

6.2.2.1.3 GFP

CCBS_N13_001 subclauses 10.1, [6] 8.3.2.1.1.2 inopportune mandatory
Ensure that the IUT in the CCBS Idle state, on receipt of a REGISTER message with a call reference in use containing a Facility information element with a CCBS-T-Request invoke component,
ignores the message and sends a STATUS message with CR2 and with a Cause information element containing the cause value #101 and remains in the same call state.

CCBS_N13_002 subclauses 10.1, [6] 8.3.2.1.1.2 invalid mandatory
Ensure that the IUT in the CCBS Idle state, with CR2 in call state N00, on receipt of a REGISTER message with CR2 containing a Facility information element with an invalid protocol profile,
sends a RELEASE COMPLETE message with CR2 containing cause #100.

CCBS_N13_003 subclauses 10.1, [6] 8.3.2.1.1.1, [8] 5.8.3.2 d inopportune mandatory
Ensure that the IUT in the CCBS Idle state, with CR2 in call state N00, on receipt of a REGISTER message with CR2, a call reference not recognized as relating to a call and with the call reference flag set to "1",
ignores the message.

CCBS_N13_004 subclauses 10.1, [6] 8.3.2.1.1.1, [8] 5.8.6.1 inopportune mandatory
Ensure that the IUT in the CCBS Idle state, with CR2 in call state N00, on receipt of a REGISTER message with CR2 which has one mandatory information element missing,
sends a RELEASE COMPLETE message with CR2 and with cause #96.

CCBS_N13_005 subclauses 10.1, [6] 8.3.2.1.1.1, [8] 5.8.6.2 invalid mandatory
Ensure that the IUT in the CCBS Idle state, with CR2 in call state N00, on receipt of a REGISTER message with CR2 and which has one mandatory information element with invalid contents,
ignores the message and sends a STATUS message with CR2 and with a Cause information element containing the cause value #100, a call state information element containing the call state and remains in the same state.

CCBS_N13_006 subclauses 10.1, [6] 8.3.2.1.2.2 inopportune mandatory
Ensure that the IUT in the CCBS Free state, with CR2 in call state N31, on receipt of a message other than FACILITY, RELEASE, RELEASE COMPLETE, STATUS or STATUS ENQUIRY with CR2,
ignores the message and sends a STATUS message with CR2 and with a Cause information element containing the cause value #101 and a call state information element containing the call state value 31.

CCBS_N13_007 subclauses 10.1, [6] 8.3.2.1.2.2 invalid mandatory
Ensure that the IUT in the CCBS Free state, with CR2 in call state N31, on receipt of a FACILITY message with CR2 containing a Facility information element with an invalid protocol profile,
ignores the message and sends a STATUS message with CR2 and with a Cause information element containing the cause value #100.

CCBS_N13_008 subclauses 10.1, [6] 8.4.2 invalid mandatory
Ensure that the IUT in the CCBS Idle state, with CR2 in call state N00, on receipt of a REGISTER message with CR2 containing a Facility information element with a CCBS-T-Request invoke component including an originatingAddress parameter of incorrect type,
ignores the (optional) originatingAddress parameter and does not reject the component with problem code of "mistyped argument".

6.2.2.2 Destination side

6.2.2.2.1 General

CCBS_N14_001 subclauses 10.2.2.1, [6] 8.3.2.1.1.1 valid optional
Ensure that the IUT in the CCBS Idle state, with CR2 in call state N00, to setup the signalling connection with the private network and to request the activation of CCBS,
sends a REGISTER message with CR2 containing a Facility information element with a CCBS-T-Request invoke component including the Bearer capability information element, destinationAddress, retentionSupported set to TRUE, and if available the High layer compatibility and Low layer compatibility information elements, and moves to call state N31 for CR2.

Selection: CCBS request retention option supported. PICS: MC 6.

CCBS_N14_002 subclauses 10.2.2.1, [6] 8.3.2.1.1.1 valid optional
Ensure that the IUT in the CCBS Idle state, with CR2 in call state N00, to setup the signalling connection with the private network and to request the activation of CCBS,

sends a REGISTER message with CR2 containing a Facility information element with a CCBS-T-Request invoke component including the Bearer capability information element, destinationAddress, retentionSupported set to FALSE, and if available the High layer compatibility and Low layer compatibility information elements, and moves to call state N31 for CR2.

Selection: CCBS request retention option NOT supported. PICS: NOT MC 6.

CCBS_N14_003 subclause 10.2.2.1 valid mandatory
Ensure that the IUT, with CR2 in call state N31, having sent a CCBS-T-Request invoke component, on receipt of a FACILITY message with CR2 containing a Facility information element with a CCBS-T-Request return result component,

takes no protocol action.

CCBS_N14_004 subclause 10.2.2.2 valid mandatory
Ensure that the IUT, with CR2 in call state N31, having sent a CCBS-T-Request invoke component, on receipt of a FACILITY message with CR2 containing a Facility information element with a reject component,

sends a RELEASE message with CR2 and cause #31.

CCBS_N14_005 subclauses 10.2.3.1, 10.2.6.1 valid mandatory
Ensure that the IUT in the CCBS Activated state, with CR1 in call state N00 and with CR2 in call state N31, on receipt of a FACILITY message with CR2 containing a Facility information element with a CCBS-T-RemoteUserFree invoke component and the IUT does not need to suspend CCBS,

sends a SETUP message with CR1 using the call establishment information used in the original call attempt and includes a Facility information element with a CCBS-T-Call invoke component and enters the call state N06 for CR1.

CCBS_N14_006 subclauses 10.2.3.1, 10.2.4.1 valid mandatory
Ensure that the IUT in the CCBS Activated state, with CR2 in call state N31, on receipt of a FACILITY message with CR2 containing a Facility information element with a CCBS-T-RemoteUserFree invoke component and the IUT does need to suspend CCBS,

sends a FACILITY message with CR2 containing a Facility information element with a CCBS-T-Suspend invoke component.

CCBS_N14_007 subclause 10.2.4.2 valid mandatory
Ensure that the IUT in the CCBS Free state, with CR2 in call state N31, having sent a FACILITY message with CR2 containing a Facility information element with a CCBS-T-Suspend invoke component, on receipt of a FACILITY message with CR2 containing a Facility information element with a reject component,

sends a RELEASE with CR2 and cause #31 and moves to call state N19 for CR2.

CCBS_N14_008 subclause 10.2.5.1 valid mandatory
Ensure that the IUT in the CCBS Free state, with CR2 in call state N31, having suspended CCBS, to request the resumption of the CCBS request,

sends a FACILITY message with CR2 containing a Facility information element with a CCBS-T-Resume invoke component.

CCBS_N14_009 subclause 10.2.5.2 valid mandatory
Ensure that the IUT in the CCBS Free state, with CR2 in call state N31, having sent a FACILITY message with CR2 containing a Facility information element with a CCBS-T-Resume invoke component, on receipt of a FACILITY message with CR2 containing a Facility information element with a reject component,

sends a RELEASE with CR2 and cause #31 and moves to call state N19 for CR2.

CCBS_N14_010 subclause 10.2.6.1 valid mandatory
Ensure that the IUT in the CCBS Free state, with CR1 in call state N00 and with CR2 in call state N31, to initiate establishment of the CCBS call,

sends a SETUP with CR1 and with the Bearer capability of the original call attempt and a Facility information element with a CCBS-T-Call invoke component and moves to call state N06 for CR1.

CCBS_N14_011 subclause 10.2.6.2 valid mandatory
Ensure that the IUT in the CCBS Free state, with CR1 in call state N01 and with CR2 in call state N31, having sent a SETUP with CR1 and with a CCBS-T-Call invoke component, on receipt of a FACILITY message with CR1 containing a Facility information element containing a reject component,
takes no protocol action.

CCBS_N14_012 subclause 10.2.6.2 valid mandatory
Ensure that the IUT in the CCBS Free state, with CR1 in call state N01 and with CR2 in call state N31, having sent a SETUP with CR1 with a CCBS-T-Call invoke component, on receipt of a RELEASE COMPLETE message with CR1 with cause indicating call failed before reaching destination,
sends a RELEASE with CR2 and enters N19 for CR2 and N00 for CR1.

CCBS_N14_013 subclause 10.2.7.1 valid mandatory
Ensure that the IUT in the CCBS Activated state, with CR2 in call state N31, in order to deactivate the CCBS request,
sends a RELEASE message with CR2 and cause #31.

6.2.2.2.2 Timers

CCBS_N15_001 subclause 10.2.6.2 timer mandatory
Ensure that the IUT in the CCBS Activated state, with CR1 in call state N10 and with CR2 in call state N31, on expiry of T-CCBS5,
sends a RELEASE message with CR2 and enters the Release Request state (N19).

6.2.2.2.3 GFP

CCBS_N16_001 subclauses 10.2, [6] 8.3.2.1.2.2 inopportune mandatory
Ensure that the IUT in the CCBS Activated state, with CR2 in call state N31, on receipt of a message other than FACILITY, RELEASE, RELEASE COMPLETE, STATUS or STATUS ENQUIRY with CR2,
ignores the message and sends a STATUS message with CR2 and with a Cause information element containing the cause value #101 and a call state information element containing the call state value 31.

CCBS_N16_002 subclauses 10.2, [6] 8.3.2.1.2.2 invalid mandatory
Ensure that the IUT in the CCBS Activated state, with CR2 in call state N31, on receipt of a FACILITY message with CR2 containing a Facility information element with an invalid protocol profile,
ignores the message and sends a STATUS message with CR2 and with a Cause information element containing the cause value #100.

7 Compliance

An ATS which complies with this TSS&TP specification shall:

a) consist of a set of test cases corresponding to the set or to a subset of the TPs specified in clause 6;

b) use a TSS which is an appropriate subset of the whole of the TSS specified in clause 5;

c) use the same naming conventions for the test groups and test cases;

d) maintain the relationship specified in clause 6 between the test groups and TPs and the entries in the PICS proforma to be used for test case deselection;

e) comply with ISO/IEC 9646-2 [4].

In the case of a) or b) above, a subset shall be used only where a particular Abstract Test Method (ATM) makes some TPs untestable. All testable TPs from clause 6 shall be included in a compliant ATS.

8 Requirements for a comprehensive testing service

As a minimum the Remote test method, as specified in ISO/IEC 9646-2 [4], shall be used by any organization claiming to provide a comprehensive testing service for network equipment claiming conformance to EN 300 359-1 [1].

Annex A (informative): Changes with respect to the previous ETS 300 359-5

The following changes have been done:

- conversion to EN layout;
- replacement of references to ETS 300 102 with EN 300 403;
- substitution of non-specific references to basic standards where the intention is to refer to the latest version.

History

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